



ST. XAVIER'S COLLEGE

Maitighar, Kathmandu

First Terminal Examination- 2023

Grade: XI

Subject: Chemistry (3011)

Time: 2 Hours

Full Marks: 50

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Choose the correct option.

[9 x 1 = 9]

- If three electrons are lost by a metal ion (M^{3+}), its oxidation number changes by
(a) -3 (b) +3 (c) -6 (d) +6
- How many electrons are needed to balance the following equation?
 $SO_3^{2-} \rightarrow SO_4^{2-} + e^-$
(a) 0 (b) 2 (c) 4 (d) 6
- The compound which acts as both oxidizing agent and reducing agent is
(a) P_2O_5 (b) H_2S (c) HN_3 (d) SO_3
- A mixture of sand and iodine is separated by
(a) crystallization (b) distillation (c) filtration (d) sublimation
- The percentage composition of water is
(a) 11.11% of H, 89.89% of O (b) 66.66% of H, 33.33% of O
(c) 98% of H, 2% of O (d) 2% of H, 98% of O
- Which of these belong to the same period?

Element	A	B	C
Atomic number	2	10	5

- (a) B, C (b) A, B (c) C, A (d) A, B & C
- According to IUPAC, atomic mass is expressed in terms of
(a) C-12 (b) H-1 (c) O-16 (d) Cl-35
- Which periods in modern periodic table have 18 elements
(a) 1st and 2nd
(b) 2nd and 3rd
(c) 3rd and 4th
(d) 4th and 5th
- During the electrolysis of aqueous NaCl solution in presence of platinum electrodes
(a) H_2 is collected at anode
(b) Na is deposited at cathode
(c) Cl_2 is obtained at anode
(d) O_2 is collected at cathode

Group B

Short Answer Questions:

[5 x 5 = 25]

- Define the oxidation number. Assign the oxidation number and arrange them in increasing order.
[1+2+2]
(a) Iodine: I_2 , HI, HIO_4 , ICl , KIO_3
(b) Manganese: $MnCl_2$, MnO_2 , $KMnO_4$, K_3MnF_6
- Balance the following redox reaction with the oxidation number method or ion electron method. Identify oxidant and reductant in the following reactions.
[2.5+2.5=5]
(a) $Zn + NO_3^- \rightarrow Zn^{2+} + NH_4^+$ (acidic medium)
(b) $K_2Cr_2O_7 + H_2SO_4 + (COOH)_2 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + CO_2 + H_2O$
- (a) State the faraday's first law of electrolysis and show the relation between electrochemical equivalent and chemical equivalent.
[2]

(b) A metallic spoon is coated with silver by passing a current of 5 amperes through the AgNO_3 solution for 5 hours. What is the thickness of silver plating if the area of the spoon is 12cm^2 . (Density of silver is 10.5 g cm^{-3} , atomic mass of Ag 107.8) [3]

13. An element has atomic number 26,

- (a) Write its electronic configuration. [1]
- (b) Predict its block type & write two important properties of the same block. [2]
- (c) Which compound is formed when the above element in the +2 oxidation state is combined with the element Y having valence shell electronic configuration $2s^2 2p^4$. [2]

14. Explain the following statement: [5]

- (a) A cation is smaller than the parent atom.
- (b) An anion is larger than the parent atom.
- (c) Effective nuclear charge increase on moving from left to right in the periodic table.
- (d) Electron affinity of fluorine is less than that of chlorine.
- (e) On moving from left to right in the periodic table, atomic size is decreased.

Group C

Long Answer Questions:

[8 x 2 = 16]

- 15. (a) What are radicals? Define acid radicals and basic radicals of salt with examples. [1+2]
- (b) Write the chemical reaction involved to detect the presence of Cl^- ions in common salt solution. [1]
- (c) Write the molecular formula of marble and mention its qualitative and quantitative significance. [2]
- (d) What is precipitate? Give an example of a precipitation reaction. [2]
- 16. (a) Give salient features of gaseous state of matter. [2]
- (b) How does the modern periodic table explain the cause of periodicity? [1]
- (c) What are the advantages of the long form of periodic table. [2]
- (d) Compare the ionization energy of N and O. Explain any two factors affecting Ionization energy. [1+2]



ST. XAVIER'S COLLEGE

Majithar, Kathmandu

Second Terminal Examination- 2023

Grade: XI

Subject: Chemistry (3011)

Time: 3 Hours

Full Marks: 75

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Choose the correct option and write down in your answer-sheet.

[11 x 1 = 11]

- The percentage of hydrogen in water
a) 88.89% b) 2% c) 11.11% d) 1%
- Which has the smallest size?
a) Na^+ b) Mg^{2+} c) Al^{3+} d) P^{5+}
- Phosphorous has the oxidation state of +3 in
a) Phosphoric acid b) Phosphorous acid
c) Hypophosphorous acid d) Meta phosphorous acid
- Which one of the following statements is not consistent with the kinetic-molecular theory of gases?
a) Individual gas molecules are relatively far apart.
b) The actual volume of the gas molecules themselves is very small compared to the volume occupied by the gas at ordinary temperatures and pressures.
c) The average kinetic energies of different gases are different at the same temperature.
d) There is no net gain or loss of the total kinetic energy in collisions between gas molecules.
- When the product of pressure and volume is plotted against pressure for a given amount of the gas, the line obtained is
a) parallel to Y- axis b) parallel to X-axis
c) linear with positive slope d) linear with negative slope
- When formic acid is treated with conc. H_2SO_4 , the gas evolved is
a) H_2S b) CO c) SO_2 d) CO_2
- Liquid Helium is possible due to
a) Ionic bond b) covalent bond c) hydrogen bond d) Vander Waal's force
- BaO_2 is an example of
a) neutral oxide b) amphoteric oxide c) acidic oxide d) peroxide
- Which of the following is responsible for ozone layer depletion?
a) CH_4 b) CO_2 c) CF_2Cl_2 d) CHCl_3
- Sodium-glucose pump is an example of
a) Primary active transport protein b) Secondary active transport protein
c) Primary passive transport protein d) Secondary passive transport protein
- Which of the following is not include in fixed cost of production?
a) Labor b) safety c) lab service d) transport

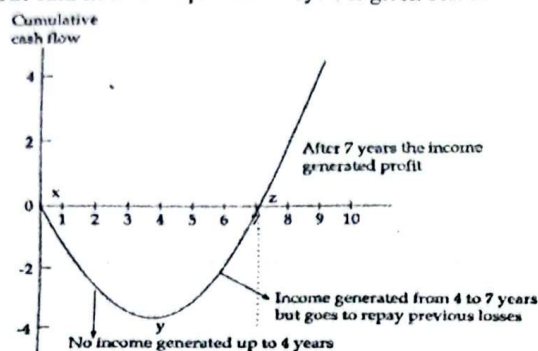
Group B

Short Answer Questions:

[8 x 5 = 40]

- a) Write any two characteristics properties of d-block elements. [2]
b) What is first ionization energy? Which compound is formed more easily among NaCl & KCl ? Give reason in terms of ionization energy of Na & K. [2]
c) Which group element in the periodic table has highest electron affinity and why? [1]
- a) Observe the following ionic reaction to answer the following question
 $\text{Cr}_2\text{O}_7^{2-} + \text{Fe}^{++} \rightarrow \text{Fe}^{3+} + \text{Cr}^{3+}$
i) Identify oxidant and reductant in the reaction. [1]
ii) Balance above redox reaction by using oxidation number method or ion electron method. [2]
b) The cost of electricity required to deposit 1g of Mg is Rs.5. How much would it cost to deposit 10 g of Al? (Al = 27, Mg = 24) [2]
- a) What is passivity of iron? Write the chemical reaction involved in test of NO_3^- . [1+2]
b) What happens when conc. HNO_3 reacts with (i) Carbon (ii) Cu? [1+1]
- a) Give a reaction to show that nascent hydrogen is strong reducing agent than molecular hydrogen. [2]
b) Hydrogen fuel does not produce pollution and effective than methane, coal and petrol. Justify. [2]
c) Write one biological effect of heavy water. [1]

16. The electronic configuration of element X is $3s^2 3p^1$ and that of Y is $3s^2 3p^4$.
- Write the name of X and Y. [1]
 - State and explain the type of bond they form with appropriate reason. [1]
 - Draw the Lewis dot structure of the compound. [1]
 - Write an example of a compound having expanded octet in its central atom and also write its Lewis dot structure. [2]
17. a) Boyle's law was put forward by Anglo-Irish chemist Robert Boyle in the year 1662. It explains relationship between the pressure and volume of the gas.
- State Boyle's law and prove, $PV = K$. [2]
 - Mountaineers carry oxygen gas cylinders. Explain in the light of Boyle's law. [1]
- b) 3 liters of an ideal gas at 780 mm Hg pressure is expanded so that the pressure of the gas becomes one third of its original pressure at constant temperature. Find the new volume of the gas. [2]
18. The cash flow in the production cycle is given below.



- What is meant by patent and copy right? [2]
 - The graph reaches minimum (y) at four years. For what purpose this time is used? [1]
 - Explain the negative cash flow in the production cycle from the figure above. [2]
19. Give reason
- Why are halogen never found in free state in nature? [1]
 - Why halides are found in sea water? [1]
 - Cl_2 added to swimming pool. Why? [1]
 - HBr cannot be prepared by using NaBr with conc. H_2SO_4 . Why? [1]
 - Write the action of bromine on hot and conc. NaOH solution. [1]

Group C

Long Answer Questions:

[3 x 8 = 24]

20. a) Give reasons for the following:

[1+1+1+1]

- H_2O is a liquid whereas H_2S is gas.
- The density of ice is less than water.
- Metal are good conductor of heat and electricity.
- Ethyl alcohol is a covalent compound yet it is completely miscible in water.

- b) Draw the Lewis structure of following.

[1+1+1+1]

- NH_4Cl
- H_2SO_4
- H_2O_2
- H_3O^+

21. Rutherford and his coworkers performed an α -particle scattering experiment to discover nucleus.

- What are α -particle? [1]
- Why Gold foil is used in this experiment? [1]
- What experimental evidence led Rutherford's to conclude that, [3]
 - The atom consists of mostly empty space.
 - The size of nucleus is extremely small as compared to size of an atom.
 - The nucleus of an atom is positively charged.
- How does Bohr's atomic model overcome the limitation of Rutherford's atomic model? [3]

22. H_2SO_4 is mainly prepared by Contact process.

- Explain the necessary condition for the preparation of high yield of H_2SO_4 . [2]
- Draw the flowsheet diagram of manufacture of H_2SO_4 . [3]
- Sulfuric acid act as dehydrating agent. Explain with a suitable reaction. [1]
- Write analytical use of H_2S gas? [2]

ST. XAVIER'S COLLEGE

Maitighar, Kathmandu

Grade: XI

Pre-annual Examination - 2024

Subject: Chemistry (3011)

Time: 3 Hours

Full Marks: 75

Candidates are required to give their answer in their own words as far as practicable.
Attempt all questions.

Group A

Write the correct option. [11 x 1 = 11]

- Formation of CO and CO₂ illustrates the law of
(a) Law of conservation of mass (b) Law of Reciprocal proportion
(c) Law of Constant Proportion (d) Law of Multiple Proportion
- Atomic spectrum of Hydrogen falling in the visible region
(a) Lyman series (b) Balmer series (c) Paschen series (d) P-funds series
- How many atoms are there in two molecules of water?
(a) 3 (b) 4 (c) 5 (d) 6
- The ratio of sigma and pi bonds in benzene is
(a) 2 (b) 4 (c) 6 (d) 8
- Deliquescence would not occur if air did not contain
(a) N₂ (b) O₂ (c) CO₂ (d) water vapor
- Which of the following species contain active nitrogen
(a) NO₂ (b) N₂O (c) N₂O₃ (d) N₂O₅
- Metal containing its own oxide as impurity can be purified by
(a) Filtration (b) Distillation
(c) Electrorefining (d) Poling
- Which of the following is not related to the Batch process?
(a) Requires high-cost equipment (b) Can not be controlled easily
(c) Generally available in fully automated plant (d) Involves a sequence of steps followed in specific order
- Alkanes are commonly called
(a) Fatty acid (b) Olefins (c) Paraffins (d) Arenes
- Stability of the carbanion is in the order of
(a) 1° > 2° > 3° (b) 1° < 2° < 3° (c) 1° > 3° > 2° (d) 3° > 1° > 2°
- Addition of HBr in to the alkene follows
(a) Nucleophilic addition (b) Electrophilic addition (c) Free radical addition (d) Electrophilic substitution

Group B

Short Answer Questions. [8x5 = 40]

- (a) State Heisenberg uncertainty Principle? [1]
(b) What is quantum number? Write down all possible values of n, m and l for the 15th electron of chlorine in ground state. [1+1]
(c) Define degenerate orbital with example. [1]
(d) Define Hund's rule for maximum multiplicity. [1]

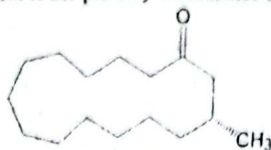
OR

- (a) What is electron affinity? Write down chemical equation to show first electron affinity of chlorine. [2]
(b) Metal and nonmetal normally form ionic bond but two nonmetal form covalent. Explain on the basis of electronegativity. [1]
(c) The first ionization energy of alkaline earth metals is smaller than the second ionization energy but these metals show bivalency in most of their compounds. Explain. [2]
- (a) $\text{NaOH} + \text{Cl}_2 \rightarrow \text{NaCl} + \text{NaClO}_3 + \text{H}_2\text{O}$
Balance the above equation by oxidation number or ion-electron method. [2]
(b) Show your conversancy with attackable electrodes. [1]
(c) Define the term electrochemical equivalent (ECE). Derive its relation with chemical equivalent. [1+1]
- (a) State law of mass action. [1]
(b) Caustic soda dissolves more when the beaker containing its solution is cooled under tap water. Why? [1]
(c) While opening the coca-cola bottle, the gas evolves. What is the principle behind this phenomenon? How? [1]
(d) Write the effect of concentration and pressure on the following equilibrium reaction. [2]
$$\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$$

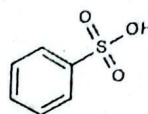
OR

- (a) State Charles' Law. [1]
(b) A gas occupies a volume of 250 cc at 700 mm pressure and 25 °C. What additional pressure is required to reduce the gas volume to its 4/5th volume at the same temperature? [3]
(c) Evaporation of liquid causes cooling of the surface. Why? [1]
- (a) Draw Lewis structure of KNO₃ and CO₃²⁻ molecules. [1+1]
(b) The bond angle at the central atom in NF₃ is 103° whereas in BF₃ is 120°. What factors account for the difference in bond angle? [2]
(c) Ammonia has a higher boiling point than phosphine. Give a reason. [1]
- Depending upon the nature of minerals present in the ores, calcination and roasting are mainly used for conversion of ores into their respective oxides.

- (a) What do you mean by roasting and calcination in the metallurgical process? [2]
 (b) Name the vessel in which roasting is carried out. [1]
 (c) Define slag. [1]
 (d) Compare the solubility of the following. [1]
 $Mg(OH)_2$, $Ca(OH)_2$, $Sr(OH)_2$, $Ba(OH)_2$
- 6) Muscone is an organic compound that is the primary contributor to the odor of musk deer. The structure of muscone is



- (a) Write the structure of the first member in the homologous series of functional group present in the above compound. [1]
 (b) Define the term [1+1]
 i) Knocking property of fuel
 ii) Gasoline additive
 (c) Write the IUPAC name of the following compounds. [1+1]
- i) $(CH_3)_2C=CHCH(OH)COOH$ ii)
- 7) (a) Differentiate homolytic and heterolytic bond fission. [2]
 (b) Ethanoic acid is a weaker acid than 2-Nitroethanoic acid, Why? [1]
 (c) How can you detect heteroelement of the following compound by sodium nitroprusside test? [1]



- (c) Write the functional isomers of molecular formula $C_2H_4O_2$. [1]
 8) (a) Write the action of HI on $CuSO_4$ solution. [1]
 (b) Arrange the following in the decreasing order of acid strength (in water) with explanation. [2]
 HCl , HF , HI , HBr
 (c) Write the formula and uses of the Magnesia. [1]
 (d) Name any two allotropes of phosphorus. [1]

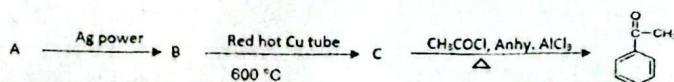
Group C

Long Answer Questions. [3x8 = 24]

- 9) (a) A person adds 1.71g of sugar ($C_{12}H_{22}O_{11}$) in order to sweeten his tea. Calculate the number of carbon atoms added? [2]
 (b) A signature written in a pencil weighs 1 mg. Calculate the number of carbon atoms present in the signature? [1]
 (c) $Ca(OH)_2 + 2NH_4Cl \rightarrow CaCl_2 + 2NH_3 + 2H_2O$
 The reaction is carried out by mixing 7g of pure $Ca(OH)_2$ and 7g of pure NH_4Cl .
 i) Find the limiting reactant. [1]
 ii) Calculate the mole of unreacted reactant left over? [1]
 iii) How many moles of $CaCl_2$ are formed? [1]
 iv) What volume of NH_3 is produced at $27^\circ C$ and 1.5 atm pressure? [2]
- 10) (a) What happens when H_2S is passed through acidified Potassium permanganate solution? [1]
 (b) Prove that conc. H_2SO_4 acts as an oxidizing agent. [1]
 (c) What are the harmful effects of ammonia? [1]
 (d) What happens when conc. nitric acid is treated with copper metal? [1]
 (e) What are the industrial and medical applications of oxygen? [2]
 (f) Write the structural features of fullerene. [1]
 (g) List down the main role of magnesium in the biological system. [1]

OR

- (a) Describe the principle for the manufacture of sodium hydroxide with a flow-sheet diagram by Diaphragm cell. [2+2]
 (b) Why is the chemical industry important? [2]
 (c) What is NPK fertilizer? [1]
 (d) What is oleum? How is it converted into sulphuric acid? [1]
- 11) (a) An alkene 'A' undergoes ozonolysis and gives compounds B and C of molecular formula C_3H_6O . B and C are functional isomers of each other. Identify the compounds A, B and C with related reactions. [3]
 (b) Identify A, B and C in the following reaction sequence. [3]



- (c) Is the following anion aromatic? [1]



- (d) What happens when ethyne is treated with HBr ? [1]



ST. XAVIER'S COLLEGE

Maitighar, Kathmandu

Annual Examination - 2024

Grade: XI

Subject: Chemistry (3011)

Time: 3 Hours

Full Marks: 75

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Choose the correct option:

[11 x 1 = 11]

- The empirical formula of ethyl alcohol is
(a) C_2H_5OH (b) CHO (c) C_2H_6O (d) CH_3CH_2OH
- One molecule of Ascorbic acid ($C_6H_8O_6$) contains mass in grams is
(a) 1.9×10^{-22} (b) 2.9×10^{-22} (c) 4.9×10^{-22} (d) 6.9×10^{-22}
- Which of the following is true in terms of size?
(a) $Na^+ > Mg^{2+} > Al^{3+} > Si^{4+}$ (b) $Mg^{2+} > Al^{3+} > Si^{4+} > Na^+$
(c) $Si^{4+} > Al^{3+} > Mg^{2+} > Na^+$ (d) $Al^{3+} > Si^{4+} > Mg^{2+} > Na^+$
- The highest dipole moment is present in
(a) CO_2 (b) CCl_4 (c) $CHCl_3$ (d) H_2
- Which of the following properties of liquids is related to internal resistance to its flow?
(a) Refractive index (b) Boiling point (c) Surface tension (d) Viscosity
- Charring of sugar by conc. H_2SO_4 is its
(a) Oxidizing action (b) Dehydrogenation action
(c) Reducing action (d) Dehydrating action
- Which of the following sulphate is least soluble in water?
(a) $SrSO_4$ (b) $BaSO_4$ (c) $CaSO_4$ (d) $BeSO_4$
- The gas involved in Bhopal gas leak tragedy was
(a) Methyl isocyanate (b) H_2S
(c) Sodium isocyanide (d) Ethyl isocyanate
- Propanal and propen-1-ol are
(a) Geometrical isomers (b) Tautomers
(c) Chain isomers (d) Position isomers
- Ethyne molecule has
(a) 2 sigma bonds and 2 pi bonds (b) 2 pi bonds and 3 sigma bonds
(c) 1 pi bond and 4 sigma bonds (d) 4 sigma bonds and 2 pi bonds
- The electrophile in sulphonation of benzene is
(a) SO_3^+ (b) HSO_3^+ (c) SO_3 (d) HSO_4^+

Group B

Short Answer Questions:

[8 x 5 = 40]

- (a) How did Bohr's atomic model explain the origin of hydrogen spectra? [2]
Sketch and name various spectral series observed in the atomic spectrum of hydrogen.
(b) An atom 'A' has atomic number ($Z=29$). Calculate the total number of s-electrons of A^{++} . [1]
(c) Hund's rule is called as a rule of maximum multiplicity, why? [1]
(d) Among the given orbitals, 1p, 2s, 3f and 4d, which are not possible? [1]
- OR
- (a) Define Shielding Effect. [1]
(b) Electronic configuration for atoms of different elements are shown below:
(i) $1s^2, 2s^2$ (ii) $1s^2, 2s^2, 2p^4$ (iii) $1s^2, 2s^2, 2p^6$ (iv) $1s^2, 2s^2, 2p^6, 3s^2$ [2]

- Which electronic configuration represents the element with the largest ionization energy and why? [2]
- (c) Define the terms:
(i) Electron affinity (ii) Metallic character
2. (a) For the given chemical reaction, [2]
- $$\text{Br}_2 + \text{OH}^- \longrightarrow \text{Br}^- + \text{BrO}_3^- + \text{H}_2\text{O}$$
- Justify the above chemical equation as a disproportionation reaction.
- (b) Calculate the oxidation state of underlined iron in $\text{Fe}_4[\underline{\text{Fe}}(\text{CN})_6]_3$ and write the name of compound. [1]
- (c) For a balanced chemical equation, [2]
- $$\text{Mn}_3\text{O}_4 + 16 \text{OH}^- \longrightarrow 3\text{MnO}_4^- + 8 \text{H}_2\text{O} + 10 \text{e}^-$$
- Calculate the charge involved to oxidize 2 mol of Mn_3O_4 to MnO_4^- in alkaline medium.
3. (a) i. Write the relationship between k_p and k_c for the gaseous reactions : [1+1]
- $$\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{NO}(\text{g})$$
- ii. Predict the favourable conditions for forward reaction in the following reaction :
- $$4\text{NH}_3 + 5\text{O}_2 \rightleftharpoons 4\text{NO} + 6\text{H}_2\text{O} + \text{heat}$$
- (b) Mention hybridization in ethene molecule with an appropriate orbital picture diagram. [1]
- (c) Predict the geometry of the given molecule on the basis of VSEPR model [1]
- (i) NH_4^+ (ii) BF_3
- (d) H_2O is liquid but H_2S is gas at room temperature, why? [1]
- OR
- (a) State Boyle's law. [1]
- (b) A flask of 0.5 litre capacity was weighed after it had been evacuated. It was then filled with a gas at 740 mm pressure at 27 °C. The increase in mass of the flask was 0.8697g. Calculate the molecular mass of the gas. [2]
- (c) Liquid drops are spherical in shape; why? [1]
- (d) Define efflorescent and deliquescent with an example of each. [1]
4. (a) Write one use of isotope of hydrogen that contains mass number 1? Draw resonance structures of ozone. [1+1]
- (b) How does ammonia react with copper sulphate solution? [1]
- (c) Why can't HBr be prepared by using NaBr and conc. H_2SO_4 ? [1]
- (d) Mention the role of iron in biological systems. [1]
5. In metallurgical process, there are different steps, like concentration, calcinations, roasting, reduction and refining.
- (a) Why the concentration of ore is carried out? [1]
- (b) Mention only two points in which roasting differs from calcination. [2]
- (c) Name the type of ore concentrated by the gravity separation method. [1]
- (d) Compare the thermal stability of the following carbonate compounds: [1]
- $$\text{BeCO}_3, \text{MgCO}_3, \text{CaCO}_3 \text{ and } \text{SrCO}_3$$
6. (a) Write down the first four members of the homologous series of alcohol with its structure. [1]
- (b) Write the IUPAC name of the following compounds. [2]
- (i) $\text{CH}_3\text{-CH=CH-CH}_2\text{-CH=CH}_2$
- (ii)
- $$\text{CH}_3\text{CH}_2\text{CH}_2\text{C(=O)OCH}_2\text{CH}_3$$
- (c) Define the terms: (i) Octane number (ii) Cracking [1+1]
7. Inductive effect, one of the electronic effects, is due to the difference in electronegativity of combining atoms.
- (a) Define +I (positive inductive effect) & -I (Negative inductive effect) [1]

- (b) Classify following as electrophile, nucleophile & free radicals: [2]
 i. H_2O ii. $\text{Cl}^\ominus$ iii. NH_3 iv. AlCl_3
- (c) Explain the following facts with reasons. [2]
 i. Chloroethanoic acid more acidic than ethanoic acid
 ii. Alcohol and Ether are functional isomers.
8. (a) Write down the action of Cl_2 on (i) cold and dilute NaOH solution (ii) H_2S [2]
 (b) What happens when carbon monoxide is treated with chlorine in the presence of sunlight? [1]
 (c) How would you test the presence of NO_3^- ion in solution? [1]
 (d) Write one use of phosphine gas. [1]

Group C

Long Answer Questions:

[8 x 3 = 24]

1. (a) State the law of multiple proportion. [1]
 (b) If it takes one second to count two rice grains, calculate the time (in years) required to count a mole of rice grains. [2]
 (c) At high temperatures, sulphurs combine with iron to form the brown-black iron (II) sulphide

$$\text{Fe (s)} + \text{S (l)} \longrightarrow \text{FeS (s)}$$

 In one experiment, 7.62 grams of Fe are allowed to react with 8.67 grams of S,
 i. Find which one is limiting the reactant and why? [1]
 ii. Calculate the mass of FeS formed. [1]
 iii. How much is excess reagent left at the end of the reaction? [1]
 (d) State Avogadro's hypothesis. Calculate the percent of water of crystallization in gypsum salt ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) [2]
2. (a) (i) Define allotropes and write different allotropic forms of carbon. [2]
 (ii) Write the action of conc. H_2SO_4 on H_2S [1]
 (b) Distinguish between batch process and continuous process. [2]
 (c) Classify the following oxides: [2]
 (i) NO_2 (ii) BaO (iii) Cr_2O_3 (iv) Na_2O_2
 (c) Why does ammonia turn mercurous nitrate paper black? [1]

OR

- (a) Describe the principle for the manufacture of sulphuric acid with a flow-sheet diagram by Contact process. [3+2]
 (b) Define the term working capital. [1]
 (c) Mention two examples of each of nitrogen fertilizer and potash fertilizer. [2]
3. Alcohol [A] having molecular formula $\text{C}_2\text{H}_5\text{OH}$ undergoes dehydration with conc. H_2SO_4 at 170°C to give [B]. [B] in Ozonolysis followed by hydrolysis gives only [C].
 (a) Identify [A], [B], and [C]. [3]
 (b) [B] can also be prepared by dehydrohalogenation of haloalkane. Identify the haloalkane with the reaction involved. [1]
 (c) What happens when [B] is hydrated? [1]
 (d) Write reaction for preparation of [B] by Kolbe's reaction. [1]
 (e) Convert [B] to butane. [1]
 (f) Does [B] give Baeyer's test? If yes write the reaction. [1]



ST. XAVIER'S COLLEGE

Maitighar, Kathmandu

First Terminal Examination- 2024

Grade: XII

Subject: Chemistry (3021)

Time: 3 Hours

Full Marks: 75

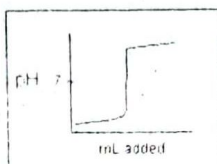
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Choose the correct option.

[11 x 1 = 11]

- 2.1 g of anhydrous barium chloride solution is completely converted into 2.36 g of barium sulphate on treating with excess sulphuric acid solution. The equivalent mass of barium (atomic mass of Ba, 137) is
a) 25.45 b) 65.46 c) 72.6 d) 42.68
- The following titration curve is the kind of curve expected for the titration of a _____ acid with a _____ base.



- (a) strong, strong (b) weak, strong (c) strong, weak (d) weak, weak
- Ostwald's dilution law is applicable to
(a) Strong electrolytes only (b) Weak electrolyte only
(c) Non-electrolytes (d) Strong and weak electrolytes
- The solubility of AgCl is minimum in
(a) Water (b) 0.1M NaCl
(c) 0.1M KCl (d) 0.1M BaCl₂
- The reaction of toluene with Cl₂ in the presence of light gives 'X'. Thus 'X' is
(a) benzyl chloride (b) o-chlorotoluene (c) m-chlorotoluene (d) p-chlorotoluene
- The colorless sweet smelling liquid compound 'A' which exposed in air forms poisonous phosgene and also react with acetone gives sleep-inducing drug. Predict the product when the compound 'A' reduced in a neutral medium?
(a) Methylene chloride (b) Methane (c) Ethyne (d) Ethane
- Aryl halides are less reactive toward nucleophilic substitution reaction than alkyl halides due to
(a) the formation of stable carbonium ion (b) resonance stabilization
(c) longer carbon-halogen bond (d) Inductive effect
- Rectified spirit is
(a) 100% pure alcohol (b) 95% ethanol + 5% water
(c) Ethanol + water + petrol (d) 95% ethanol + 55% methane
- Which of the following organometallic compound is known as anti-knocking reagent in petroleum industry?
(a) Trimethyl aluminium (b) Grignard reagent
(c) Tetraethyl lead (d) diethyl Cadmium
- What is the name of the alloy of copper called which contains a composition of 30-35% of zinc?
(a) Bronze (b) Duralumin (c) Magnalium (d) Brass
- Which of the following act as a reducing agent in the reduction zone of blast furnace during the extraction of iron from hematite?
(a) CO₂ (b) CO (c) C (d) SO₂

Group B

Short Answer Questions:

[8 x 5 = 40]

- The underlying principle of volumetric analysis is Law of chemical equivalence which states that the substances react in their equivalent proportion.
a) Derive Normality equation. [2]
b) 3.896g of soda crystal (Na₂CO₃.xH₂O) were dissolved in 250 mL of water. 20 mL of resulting solution required 21.8 mL of N/10 H₂SO₄ for exact neutralization. Calculate the percentage of anhydrous Na₂CO₃ in the crystal and the value of x. [3]
- The ionic product of water at 25°C is 1.0x10⁻¹⁴ and water is regarded as very weak electrolyte.
a) Define Weak electrolyte [1]
b) Define the ionic product of water. Deduce the relation K_w = [H₃O⁺] [OH⁻] [2]

- c) Calculate the $[H^+]$ of 0.001M NaOH solution at 25°C. [1]
 d) Explain the effect of temperature on the ionic product of water. [1]
 11. 0.49 g of pure H_2SO_4 is dissolved in 500 mL of solution.
 a) Calculate the molar concentrations of H_2O and SO_4^{2-} in the solution. [2]
 b) Calculate the pH of the solution. [1]
 c) If the above solution is diluted to 5 litre, what would be the pH of the diluted solution? [2]
 15. Compound 'A' with the molecular formula C_4H_9Br is treated with aq. KOH solution. The rate of this reaction depends upon the concentration of the compound 'A' only. When another optically active isomer 'B' of this compound was treated with aq. KOH solution, the rate of reaction was found to be dependent on concentration of compound and KOH both.
 a) Write down the structural formula of both compounds 'A' and 'B'. [2]
 b) Out of these two compounds, which one will be converted to the product with inverted configuration? [1]
 c) How is trichloromethane prepared in laboratory? Give a chemical equation for the conversion of chloroform into ethyne. [2]
 16. Give the example of each of the following reactions. [1x5=5]
 a) Oxo process
 b) Sandmeyer reaction
 c) Ozonolysis
 d) Esterification reaction
 e) Williamson's ether synthesis
 17. Chlorobenzene is used primarily as a solvent, a degreasing agent, and a chemical intermediate.
 a) Write the possible isomers for C_7H_7Cl with their IUPAC names. [2]
 b) Give the chemical equation for preparation of DDT from chlorobenzene? [1]
 c) Starting from Chlorobenzene, how will you obtain
 (i) Biphenyl (ii) Benzoic acid? [1+1]
 18. a) If you are given 1-propanol, 2-propanol and 2-methyl-2-propanol, how would you distinguish by Victor-Meyer method? [3]
 b) Starting from methyl magnesium bromide how would you prepare
 (i) 2-methyl-2-propanol (ii) ethanoic acid [1+1]
 19. White vitriol is hydrated zinc sulphate which can be used as an eye lotion.
 a) Write any two methods for the preparation of white vitriol. [1]
 b) White vitriol is efflorescent in nature. What does it mean? [1]
 c) Write down the action of heat on white vitriol. [1]
 d) How is lithopone formed? Write down its one important use. [2]

Group C

Long Answer Questions:

- [3 x 8 = 24]
 20. a) Write a balanced equation for the dissociation of the following in water and identify the conjugate acid-base pairs. [1+1+1+1]
 i) NH_4^+ ii) H_2SO_4 iii) CH_3COOH iv) H_3BO_3
 b) The solubility product of $Ca(OH)_2$ at 25°C is 4.42×10^{-5} . A 500 mL of saturated solution of $Ca(OH)_2$ is mixed with equal volume of 0.4M NaOH. Calculate the mass of $Ca(OH)_2$ precipitated out. [4]
 21. Primary standard solution can be prepared by direct weighing exact quantity of the substance and dissolving in water to make definite volume of solution. Answer the following questions.
 a) What are pre-requisites of primary standard substances? [2]
 b) Give two examples of secondary standard substance. [1]
 c) What is titration error? [1]
 d) What are molar and normal solutions? [1]
 e) Suggest two commonly used acid-base indicator with their pH range? [1]
 f) A student weighs 0.65 g of oxalic acid to make 100 mL of decinormal solution of it. Express the actual concentration using the term factor. [2]
 22. a) An alcohol 'A' ($C_4H_{10}O$) on oxidation with acidified potassium dichromate gives carboxylic acid 'B' ($C_4H_8O_2$). Compound 'A' when dehydrated with conc. H_2SO_4 gives compound 'C'. Treatment of 'C' with HBr to give 'D' then 'D' is reacted with aq. NaOH give compound 'E', which is an isomer of 'A'. Compound E is resistant to oxidation. Identify A, B, C, D and E with all the reactions involved. [1+1+1+1+1]
 b) Among Compound 'A' and 'E', Which one have highest boiling point? [1]
 c) Write the chemical reaction in which biocatalyst is used in industrial preparation of a compound known to you. [1]
 d) How will you distinguish ethanol and propan-1-ol? [1]



ST. XAVIER'S COLLEGE
Maitighar, Kathmandu
Second Terminal Examination- 2024
Grade: XII

Subject: Chemistry (3021)

Time: 3 Hours

Full Marks: 75

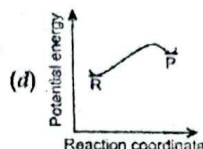
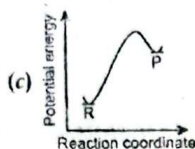
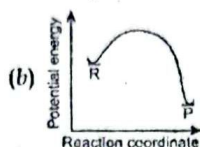
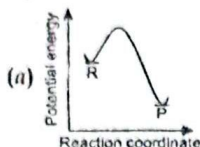
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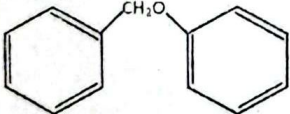
Group A

Choose the correct option.

[11 x 1 = 11]

- What is the concentration of nitrate ions if equal volume of 1M NaNO_3 and 1M KCl are mixed
a) 0.25M b) 0.5M c) 1M d) 2M
- If the solubility of A_3B_2 is s, then its solubility product will be
a) $27s^3$ b) $72s^5$ c) $108s^5$ d) $525s^5$
- An endothermic reaction with high activation energy for the forward reaction is given by the diagram.



- Which of the following statements about the First Law of Thermodynamics is correct?
a) Energy can be created but not destroyed.
b) The total energy of an isolated system is constant.
c) Work and heat are state functions.
d) The entropy of the universe always increases
- The reaction $\text{C}_6\text{H}_5\text{OH} + \text{Na} \rightarrow$ sodium phenoxide + $1/2\text{H}_2$ shows
a) Acidic nature of phenol b) basic nature of phenol
c) amphoteric nature of phenol d) reducing nature of phenol
- The primary amine is changed into corresponding alcohol by:
a) Acidified KMnO_4 b) HNO_2 c) Br_2/KOH d) HNO_3
- Which alkene on ozonolysis gives $\text{CH}_3\text{CH}_2\text{CHO}$ and CH_3COCH_3
a) $\text{CH}_3\text{CH}_2\text{CH} = \text{C} \begin{smallmatrix} \text{CH}_3 \\ \text{CH}_3 \end{smallmatrix}$ b) $\text{CH}_3\text{CH}_2\text{CH} = \text{CHCH}_2\text{CH}_3$
c) $\text{CH}_3\text{CH}_2\text{CH} = \text{CHCH}_3$ d) $\text{CH}_3 - \text{CH} = \text{CHCH}_3$
- Diethyl ether finds use in medicine is
a) Pain killer b) a hypnotic c) an antiseptic d) an anesthetic
- What products are formed when the shown ether is heated with hydrogen iodide?

a) $(\text{C}_6\text{H}_5)\text{CH}_2\text{I}$ and $(\text{C}_6\text{H}_5)\text{OH}$ c) $(\text{C}_6\text{H}_5)\text{CH}_2\text{OH}$ and $(\text{C}_6\text{H}_5)\text{I}$
b) $(\text{C}_6\text{H}_5)\text{CH}_2\text{OH}$ and $(\text{C}_6\text{H}_5)\text{CH}_2\text{I}$ d) $(\text{C}_6\text{H}_5)\text{OH}$ and $(\text{C}_6\text{H}_5)\text{I}$
- Mercury can exhibit multiple oxidation states. In which compound does mercury exist in the +1 oxidation state?
a) HgCl_2 b) Hg_2Cl_2 c) HgS d) HgO
- The ion present in Nessler's reagent is,
a) Hg^+ b) Hg^{++} c) HgI_2^{2-} d) HgI_4^{2-}

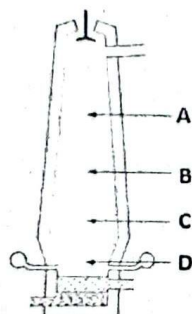
Group B

Short Answer Questions:

[8 x 5 = 40]

- What is meant by the normality factor? What volume of 95% sulphuric acid (density = 1.85 g/cc) and what mass of water must be taken to prepare 100cc of 15% solution of sulphuric acid (density 1.1g/cc)
[1+4]
- a) State Hess's law of constant heat summation.
b) At 25°C , the standard enthalpies of formation benzoic acid, $\text{CO}_2(\text{g})$, and $\text{H}_2\text{O}(\text{l})$ are -408, -393 and -286 kJmol^{-1} respectively, calculate the heat of combustion of benzoic acid i) at constant pressure and ii) at constant volume ($R = 8.314\text{Jmol}^{-1}\text{K}^{-1}$)
[1+2+2]
- F.D. Ostwald studied the relation between the degree of ionization and concentration of the solution and enunciated a law known as Ostwald's dilution law.

- a) Derive the relation $\alpha = \sqrt{\frac{K_a}{C}}$ [2]
 b) Do strong electrolytes obey this law? Explain it. [1]
 c) 1M HCN is 5.5% ionized. Calculate its ionization constant. [1]
 d) Find the pH of 0.1M CH₃COOH if it is 1.34% ionized. [1]
15. Define the instantaneous rate of reaction Compare the rate of reaction of all the components of the following reaction:
 $2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$
 Find the rate of each component in mols^{-1} , when 2.24 Liter of O₂ at NTP are produced in 30 minutes. [1+4]
16. a) Hydroboration followed by oxidation of propene gives alcohol, both steps follow Markovnikov's rule but it seems to be following anti-Markovnikov's addition of water to unsymmetrical alkene, why?
 b) Define oxonium salt. What happens when oxonium salt is dissolved in water? [3+2]
17. a) Aliphatic amines are more basic than aniline. Why? [2]
 b) Write an example of each
 i) Hoffman's hypobromite reaction [1]
 ii) Coupling reaction [1]
 iii) Rosenmund's reaction [1]
18. Steel is an alloy of iron which has versatile use in our daily life.
 a) How is steel manufactured by open hearth process? [3]
 b) Label the important zone of blast furnace in manufacture of iron.



Extraction of iron in a blast furnace.

19. X The electronic configuration of Zn is [Ar] 4s² 3d¹⁰, it is last element of 3d transition series. [2]
 a) Group 12 elements are known as pseudo-transition elements, why? [2]
 b) How is the Rinnmann's green prepared? Write it's an important application. [2]
 c) How can you prepare philosopher's wool by using zinc? [1]

Group C

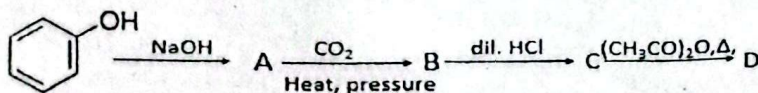
Long Answer Questions:

[3 x 8 = 24]

20. a) Define rate law. What are the effects of (i) temperature and (ii) Catalyst on rate of reaction [1+2+2]
 b) The reaction between X and Y is of 1st order w.r.t X and zero order w.r.t Y. Fill in the blanks in the following table. [3]

Expt.no	[X]molL ⁻¹	[Y]molL ⁻¹	Initial rate molL ⁻¹ min ⁻¹
1	0.1	0.1	2×10^{-2}
2	-	0.2	4×10^{-2}
3	0.4	0.4	-
4	-	0.2	2×10^{-2}

21. a) convert the following organic compounds
 i) benzene into m-bromo phenol [2+2]
 ii) Ethoxy-ethane into ethanoyl chloride [1+1+1+1]
 b) Identify A, B, C, and D.



22. Write the possible functional isomers of C₃H₉N. [1]
 a) Separate these isomers by using Hoffmann's method. [3]
 b) Represent the regular sequence of chemical reactions of the following sets of organic compounds using suitable reagents, catalysts, or conditions giving proper reasons.
 CH₃CH₂NC, CH₃CH₂Br, CH₃CH₂NH·CH₃, CH₃CH₂NH₂. [4]
